O‘QUV QO‘LLANMA · 2026 - YILKUZ

Chiziqli algebra I

1–4 haftalar

Har bir kurs fayli va tayinlangan darslik bo‘limlaridagi barcha savollar va formulalar — vektorlar, chiziqli kombinatsiyalar va span; skalyar va vektor ko‘paytmalar; matritsalar, matritsa algebrasi, determinantlar; rang va teskari matritsa — har bir masala to‘liq bosqichma-bosqich yechim bilan. Mustaqil o‘rganish va Test 1 ga tayyorgarlik uchun tuzilgan

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Manbalar: kurs konspektlari, darslar, ISA, TA sessiyalari; G. Strang, Introduction to Linear Algebra (4e) va Linear Algebra and its Applications (4e)

SAVOLLAR BAZASI · FORMULALAR KATALOGI · TO‘LIQ YECHIMLAR OKTABR 2026

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1 Kurs xaritasi va hujjatdan qanday foydalanish

Hujjat bu 2026-yil kuz semestrining birinchi choragi bo‘lgan ”Chiziqli algebra I” kursining birinchi to‘rt haftasiga mo‘ljallangan to‘liq va mustaqbil qo‘llanmadir. U kurs repositoriyasining course-files/ papkasidagi har bir tegishli manbadan qo‘lda tuzilgan: haftalik konpektlar (Notes), ma’ruzalar prezentatsiyalari, amaliy mashg‘ulotlar varaqalari, mustaq ish uch vazifalar (ISA), TA sessiyalarining masalalari va G. Strang darsliklarining belgilangan bo‘limlari. Masalalar so‘zma-so‘z takrorlangan; yechimlar to‘liq, bosqichma-bosqich, tekshirishlar bilan.

Hafta	Chiziqli algebra I
1	Vektorlar; chiziqli kombinatsiyalar; span; chiziqli mustaqillik
2	Skalyar ko‘paytma, burchaklar, proyeksiyalar; vektor ko‘paytma, yuzalar; matritsalar: amallar
3	Matritsa algebrasi; transponir; matritsalarni ko‘paytirish; determinantlar; aralash ko‘paytma
4	Matritsa rangi; Gauss-Jordan usulida teskari matritsa

Yozuv. Matritsat satr bo‘yicha yoziladi: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ — birinchi satr ab, ikkinchi cd; ustun-vektor $\begin{bmatrix} x \\ y \end{bmatrix}$, satr-vektor $\begin{bmatrix} x & y \end{bmatrix}$ ko‘rinishida yoziladi. Darajalar va indekslar ^{} va_ orqali yoziladi: A^{-1}, A^T, a_{11}.

2 Hafta 1 — Vektorlar (Chiziqli algebra I)

2.1 Chiziqli algebra I: 1-hafta formulalari katalogi

Vektorlar R^n dagi ustunlardir; 1-haftaning hammasi qo’shish, skalyarga ko’paytirish, birikmalar, span va mustaqillikdan quriladi.

$$(u+v)_i = u_i + v_i, \quad (c\,u)_i = c\,u_i \text{ (komponentalar bo’yicha amallar)} \tag{24}$$

$$c_1v_1 + \cdots + c_nv_n = \text{chiziqli birikma} \tag{25}$$

$$\text{span}\{v_1, \ldots, v_n\} = \text{bunday barcha birikmalar to’plami} \tag{26}$$

$$v_1, \ldots, v_n \text{ mustaqil} \Leftrightarrow c_1v_1 + \cdots + c_nv_n = 0 \Rightarrow c_1 = \cdots = c_n = 0 \tag{27}$$

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}, \quad \hat{x} = \frac{x}{\|x\|} \text{ (birlik vektor)} \tag{28}$$

$$u \text{ yo⁄nalishda } L \text{ uzunlikdagi } v: \; v = \frac{L\,u}{\|u\|} \tag{29}$$

$$d(P,Q) = \|Q - P\| \tag{30}$$

2.2 Chiziqli algebra I, 1-hafta: ISA mashqlari — savollar va to‘liq yechimlar

$$[][] - 2 - 1$$

Butun vektorlar bo‘limida vektorlar ustunlardir; A qismida $Au =, v =$. 3 5

$$[][] - 2 - 1$$

1. $u = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 5 \end{bmatrix}$ **bo‘lsin.** $u + v, u - v$**, 2u, −3v, 2u − 3v ni hisoblang.**

[Manba: ISA LA, hafta 1, mashq 1]

Yechim. (24) bo‘yicha komponent bo‘yicha: $u+v = \begin{bmatrix} -3 \\ 8 \end{bmatrix}$; $u-v = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$; 2u $= \begin{bmatrix} -4 \\ 6 \end{bmatrix}$; $-$3v $= \begin{bmatrix} 3 \\ -15 \end{bmatrix}$; 2u − 3v $= \begin{bmatrix} -4 & + & 3 \\ 6 & + & 15 \end{bmatrix} = \begin{bmatrix} -1 \\ 21 \end{bmatrix}$.

2. Berilgan $a = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$, $b = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$**; toping** $a + b, a - b$**, 3a + 2b.**

$$\begin{bmatrix} 5 & -1 \\ 6 & 12 \end{bmatrix}$$

[Manba: ISA LA, hafta 1, mashq 2]

Yechim. $a+b = \begin{bmatrix} -1 & + & 2 \\ 4 & & 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$; $a-b = \begin{bmatrix} -1 & & 2 \\ 4 & + & 2 \end{bmatrix} = \begin{bmatrix} -3 \\ 6 \end{bmatrix}$; 3a + 2b $= [3\cdot(-1)+2\cdot 2;$ $3 \cdot 4 + 2 \cdot (-2)] = \begin{bmatrix} 1 \\ 8 \end{bmatrix}$.

$$\begin{bmatrix} 2 & 6 & 12 \\ -4 & 8 & \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix}$$

3. Toping x**, agar** $x + \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 3 \\ -2 \end{bmatrix}$**.**

[Manba: ISA LA, hafta 1, mashq 3]

Yechim. Ma’lum vektorni ikkala tomondan ayiramiz: $x = \begin{bmatrix} 7 & & 2 \\ 3 & + & 4 \\ -2 & & 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ -3 \end{bmatrix}$.

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} - \begin{bmatrix} 2 \\ -4 \\ 1 \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \\ -2 \end{bmatrix}$$

4. Toping x**, agar 3x − 2u** $= v$**, bunda** $u = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 8 \\ 2 \end{bmatrix}$**.**

[Manba: ISA LA, hafta 1, mashq 4]

Yechim. Ajratamiz: 3x $= v+$ 2u $= \begin{bmatrix} 8 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$, shundan $x = [\frac{4}{3}$; 4/3]. Skalyarga ko'paytirish qo'shishga nisbatan taqsimlanuvchi, shuning uchun vektorni 3 ga bo'lish degani, uning har bir komponentini 3 ga bo'lishdir.

5. Vektorlarning Evklid uzunligini hisoblang: $u = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $v = \begin{bmatrix} 2 \\ -3 \\ 6 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

[Manba: ISA LA, hafta 1, mashq 5]

Yechim. (28) ga ko'ra: $\|u\| = \sqrt{9} + 16 = 5$; $\|v\| = \sqrt{4} + 9 + 36 = 7$; $\|w\| = 0$ (faqat nol vektor nol uzunlikka ega).

6. $v = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ **vektoriga bir xil yo'nalishda birlik vektorni toping va uning uzunligi 1 ga teng ekanini tekshiring.**

$$[-\frac{4}{5}; \frac{3}{5}]$$

[Manba: ISA LA, hafta 1, mashq 6]

Yechim. $\|v\| = \sqrt{16} + 9 = 5$, shuning uchun $\hat{v} = v/5 = [-\frac{4}{5}$; 3/5]. Tekshirish: $\|\hat{v}\| = \sqrt{\frac{16}{25} + \frac{9}{25}} = 1$.

7. $v = \begin{bmatrix} 3 \\ -8 \end{bmatrix}$ **vektoriga bir xil yo'nalishda uzunligi 10 bo'lgan vektorni toping.**

[Manba: ISA LA, hafta 1, mashq 7]

Yechim. Bu yerda $\|v\| = 5$, shuning uchun $\frac{10}{5} = 2$ miqdorida masshtablaymiz: javob 2v =

$$\begin{bmatrix} 6 \\ -16 \end{bmatrix}$$

‖2v‖$= 2 \cdot 5 = 10$ norma bir xilligi bo'yicha.

8. Vektorlarni ayirish va uzunlikdan foydalanib, $P = (1, -2, 3)$ **va** $Q = (5, 1, -3)$ **orasidagi masofani toping.**

[Manba: ISA LA, hafta 1, mashq 8]

Yechim. $PQ = Q - P = \begin{bmatrix} 5 \\ 1 \\ -3 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$, shuning uchun $d(P, Q) = \|PQ\| = \sqrt{16} + 9 + 36 = \sqrt{61}$.

$$[5 - 1; 1 - (-2); -3 - 3]$$

$$\begin{bmatrix} 4 \\ 3 \\ -6 \end{bmatrix}$$

. Shuning uchun $d(P,\,Q) = \|PQ\| = \sqrt{4^2} + 3^2 + (-6)^2 = \sqrt{16} + 9 + 36 = \sqrt{61}$.

9. $v_1 =, v_2 =$ **bo’lsin.** $2v_1 + v_2,\ -v_1 + 3v_2,\ 4v_1 - 2v_2,\ 0v_1 + 5v_2$ **ni hisoblang. 2 3**

$$[][][][][]1 - 1 - 1 - 24 + 26$$

[Manba: ISA LA, hafta 1, mashq 9]

Yechim. $2v_1 + v_2 =$

$$\begin{bmatrix} 2 & 2 \\ 6 & 3 \end{bmatrix}$$

; $-v_1 + 3v_2 =$

$$\begin{bmatrix} -1 & 3 \\ -3 & 9 \end{bmatrix}$$

; $4v_1 - 2v_2 =$

$$\begin{bmatrix} 4 & -2 \\ 12 & -6 \end{bmatrix}$$

; $0v_1 + 5v_2 =$ (nol skalyar v_1 ni to’liq yo’q qiladi).

$$\begin{bmatrix} 0 & 5 \\ 0 & 15 \end{bmatrix}$$

$5v_2 =$

$$\begin{bmatrix} 1 & -1 & 7 \\ -1 & -2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 6 \\ 0 & 0 \end{bmatrix}$$

.

10. c_1, c_2 **ni toping shundayki** $c_1 + c_2 =$**; almashtirish orqali tekshiring. 1 2 1**

[Manba: ISA LA, hafta 1, mashq 10]

Yechim. Komponentli tenglamalar: $c_1 - c_2 = 7$ va $c_1 + 2c_2 = 1$. Birinchisini ikkinchisidan ayiramiz:

$$c_1 + c_2 = 1; c_1 - c_2 = 7; 2c_1 + c_2 = 1.$$

$3c_2 = -6$, demak $c_2 = -2$ va $c_1 = 5$. Tekshirish:

$$\begin{bmatrix} 5 & & 2 \\ 5 & + & 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$$

.

11. $b = \begin{bmatrix}5\\1\\4\end{bmatrix}$ **vektorini** $b = c1 \cdot v1 + c2 \cdot v2 + c3 \cdot v3$ **ko'rinishida yozish mumkinmi, bunda** $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\0\end{bmatrix}$**?**

[Manba: ISA LA, hafta 1, mashq 11]

Yechim. Komponentlar $c_1 + c_3 = 5$, $c_2 + c_3 = 1$, $c_1 + c_2 = 4$ beradi. Birinchi ikkitasini qo'shib, uchinchisini ayiramiz: $2c_3 = 2$, demak $c_3 = 1, c_1 = 4, c_2 = 0$. Kombinatsiya mavjud:$b = 4v_1 + 0 \cdot v_2 + 1 \cdot v_3$. Yechimning noyobligi shuni ham ko'rsatadiki, uchta vektor chiziqli bog'liq emas.

12. $c \cdot \begin{bmatrix}-2\\3\end{bmatrix}$, $c \in R$ **bo'lgan barcha vektorlarni geometrik tasvirlang;** $\begin{bmatrix}4\\-6\end{bmatrix}$, $\begin{bmatrix}-2\\3\end{bmatrix}$, $\begin{bmatrix}6\\-8\end{bmatrix}$ **vektorlari bu** span **ga tegishlimi?**

[Manba: ISA LA, hafta 1, mashq 12]

Yechim. Span — boshlang'ich nuqtadan o'tuvchi (−2, 3) yo'nalishidagi butun to'g'ri chiziq. Tekshirish: $\begin{bmatrix}4\\-6\end{bmatrix}$ — ha ($c = -2 : -2 \cdot (-2) = 4$ va $-2 \cdot 3 = -6$); $\begin{bmatrix}-2\\3\end{bmatrix}$ — ha ($c = 1$); $\begin{bmatrix}6\\-8\end{bmatrix}$ uchun −2c = 6 va 3c = −8, ya'ni $c = -3$ va $c = -\frac{8}{3}$ — ziddiyat, demak u to'g'ri chiziqda yotmaydi.

13. $v_1 =, v_2 =$ **vektorlari chiziqli mustaqilmi?** $c_1v_1 + 26c_2v_2 = 0$ **bo'lgan nol bo'lmagan** c_1, c_2 **ni toping va geometrik tushuntiring.**

[Manba: ISA LA, hafta 1, mashq 13]

Yechim. $v_2 = 3v_1$, shuning uchun $3v_1 - v_2 = 0$ — no trivial bog'liqlik ($c_1 = 3, c_2 = -1$). Vektorlar bog'liq. Geometrik jihatdan ikkalasi ham boshlang'ich nuqtadan o'tuvchi bir to'g'ri chiziqda yotadi, shuning uchun ular birga hali ham faqat shu to'g'ri chiziqni hosil qiladi — bog'liqlik bitta vektor yangi yo'nalish bermasligini anglatadi.

$$\begin{bmatrix}1 & 0\\1 & \end{bmatrix}$$

14. $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\2\end{bmatrix}$ **vektorlari chiziqli mustaqilmi? Agar bog'liq bo'lsa, birini boshqalari orqali ifodalang.**

[Manba: ISA LA, hafta 1, mashq 14]

Yechim. $v_3 = \alpha v_1 + \beta v_2$ deb faraz qilamiz: komponentlar $\alpha = 1$ (birinchi), $\beta = 1$(ikkinchi), $\alpha+\beta = 2$(uchinchi) — mos keladi. Demakv$_3 = v_1+v_2$, no trivial bog'liqlik mavjud va to'plam bog'liq. Faqat ikkita mustaqil yo'nal mavjud; span — tekislik.

$$\begin{bmatrix}1\\1\\1\end{bmatrix}$$

15. $v_1 =, v_2 =$ **bo’lsin. (1) Ular** R^2 **ni hosil qiladimi?** $(2)b =$ **uchun**

$$21[]b_27$$

c_1, c_2 **ni toping shundayki** $c_1v_1 + c_2v_2 = b$**. (3) formuladan ifodalang. 4**

[Manba: ISA LA, hafta 1, mashq 15]

Yechim. $(1) - (2)c_1 + 3c_2 = b_1,\ 2c_1 + c_2 = b_2$ tenglamalarini yechamiz. Chiqarib: $c_1 = b_1 - 3c_2, u$ holda $2b_1 - 6c_2 + c_2 = b_2$, demak $c_2 = (2b_1 - b_2)/5$ va $c_1 = (-b_1 + 3b_2)/5$. Formulalar har qanday (b_1, b_2) uchun aniqlangan, shuning uchun ha, ikkita vektor R^2 ni hosil qiladi. $(3)c_1 = (-7 + 12)/5 = 1$, $c_2 = (14 - 4)/5 = 2$: $b = v_1 + 2v_2$.

$$Q = uv = (2 : a = V) + 2Vo.$$

16. $u_1 =, u_2 =$ **bo’lsin. (1) Ular** R^2 **ni hosil qiladimi? (2) 2 3**

$$(1)u_2 = 6u_1, po$$

span**{**u_1, u_2 **} ni geometrik ravishda tasvirlang; (3) unga tegishlimi? −4 2**

[Manba: ISA LA, hafta 1, mashq 16]

Yechim. Nisbatlarni solishtirish $-6/(-1) = 6$ va 3/2 ko’rsatadiki, bosilgan juftlik proporsional emas, shuning uchun bosilgandek uR^2 ni qamrab oladi; mashqning standart versiyasida $u2 = 3u1 = \begin{bmatrix}-3\\6\end{bmatrix}$ — uni yechamiz. $(1)u2 = 3u1$, shuning uchun span — faqat span$\{u1\}$. (2) Geometrik jihatdan bu (−1, 2) yo’nalishidagi boshidan o’tgan yagona to’g’ri chiziq; ikki kollinear vektor hech qachon tekislikni qamrab olmaydi. $(3)\begin{bmatrix}8\\-4\end{bmatrix}$ uchun: $-c = 8$ va 2c $= -4$, ya’ni $c = -8$ va $c = -2$ — ziddiyat, demak yo’q; $\begin{bmatrix}1\\2\end{bmatrix}$ uchun: $-c = 1$ va 2c $= 2$, ya’ni $c = -1$ va $c = 1$ — ziddiyat, demak yo’q.

$$y = (\frac{1}{2})\begin{bmatrix}1 & 0\\1 & 1\end{bmatrix}x + \begin{bmatrix}1\\0\end{bmatrix}(16)$$

17. Faraz qilaylik $v1 = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v2 = \begin{bmatrix}0\\1\\1\end{bmatrix}$, $v3 = \begin{bmatrix}1\\1\\0\end{bmatrix}$**.** $(1)c1v1 + c2v2 + c3v3 = b$ **ni ixtiyoriy** $b = \begin{bmatrix}b1\\b2\\b3\end{bmatrix}$ **uchun yeching; (2) nima uchun bu uch vektor** R^3 **ni qamrab olishini**

tushuntiring; $(3) b = \begin{bmatrix} 4 \\ 2 \\ 6 \end{bmatrix}$ **ni ifodalang.**

$$c_1 = (b_1 + b_2 - b_3)/2$$

ixtiyoriyb $= b_2$ **uchun yeching; (2) uchta vektor** R^3 **ni egallaydigan sababini tushuntiring;**

$$c_1 = (b_1 + b_2 - b_3)/2$$

[Manba: ISA LA, hafta 1, mashq 17]

Yechim. (1) Komponentlar: $c_1+c_3 = b_1$, $c_2+c_3 = b_2$, $c_1+c_2 = b_3$. Uchtasini qo'shamiz: $2(c_1 + c_2 + c_3) = b_1 + b_2 + b_3$, demak $c_1 + c_2 + c_3 = {}^{b1+b2+b3}$. Keyin $c_1 = {}^{b1-b2+b3}$, $22c_2 = {}^{-b1+b2+b3}$, $c_3 = {}^{b1+b2-b3}$. (2) Yechim istalgan o'ng tomon uchun mavjud va yagona, shuning uchun span — butun R^3 (vektorlar mustaqil). $(3) c_1 = {}^{4-2+6} = 24$, $c_2 = {}^{-4+2+6} = 2$, $c_3 = {}^{4+2-6} = 0$:$b = 4v_1 + 2v_2$. 2 2

$$b = 4v_1 + 2v_2$$

18. Detalga uchta kuch ta'sir qiladi: $F_1 = N, F_2 = N, F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

N**. (1) Rivojlantiruvchini toping; (2) detali muvozanatga keltiruvchi**F_4 **ni toping:**$F_1 + F_2 + F_3 + F_4 = 0$**.**

$$\begin{bmatrix} 40 \\ -15 \\ -20 \end{bmatrix}$$

[Manba: ISA LA, hafta 1, mashq 18]

Yechim. (1) Komponent bo'yicha qo'shamiz: $F_1 + F_2 + F_3 =$

$$\begin{bmatrix} 10 \\ 25 \\ -5 \end{bmatrix}$$

$= 50N$. (2) Muvozanat uchun $F_4 = -(F_1 + F_2 + F_3) =$

$$\begin{bmatrix} -40 \\ 15 \\ 20 \end{bmatrix}$$

N kerak: bu rivojlantiruvchi vektorga aynan qarama-qarshi, shunda umumiy kuch nolga aylanadi.

2.3 Chiziqli algebra I, hafta 1: TA sessiya masalalari (alternativ to'plam) — savollar va to'liq yechimlar

TA to'plami sakkizta qisqa masaladan iborat, har biri rasmiy yechimga ega; ikkalasini ham keltiramiz, taxminlarni batafsil ochib berib.

1. $a = (-1, 3, 2), b = (4, 0, -2)$**. Hisoblanga** $+b$ **va 3a** $-b$**;** a **vab** orasidagi o'rtacheni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 1]

Yechim. $a + b = (3, 3, 0)$; 3a $-b = (-3 - 4,\ 9 - 0,\ 6 + 2) = (-7,\ 9,\ 8)$. O'rta — o'rtacha $^{a+b} = (\frac{3}{2}, 3_2, 0)$ — bu kesma o'rtasining skalyar formuladagi o'rtacha olish g'oyasi bilan bir xil.

2. Berilganp $+q = (5, 7)$ **vap** $-q = (-1, 3)$**;**p **vaq ni toping.**

[Manba: Zadachi TA, nedelya 1, zadacha 2]

Yechim. Ikkita tenglamani qo'shamiz: 2p $= (4, 10)$, demakp $= (2, 5)$. Ayiramiz: 2q $= (6, 4)$, demakq $= (3, 2)$. Tekshirish: $(2, 5) + (3, 2) = (5, 7)$ va $(2, 5) - (3, 2) = (-1, 3)$. «Qo'shib ayirish» usuli — bu ikkita skalyar tenglamani bir vaqtda yechish vektor analogi.

3. au $= (1,\ 2,\ -1), v = (2,\ -1,\ 0)$**. Hisoblang** $-u+$ **3v; yozing cu + dv komponent bo'yicha; cu** $+dv(5, 0, -1)$ **ga teng bo'lishi mumkinmi?**

[Manba: Zadachi TA, nedelya 1, zadacha 3]

Yechim. $-u+$ 3v $= (-1,\ -2,\ 1) + (6,\ -3,\ 0) = (5,\ -5,\ 1)$. Komponent bo'yicha cu $+dv = (c+$ 2d, 2c $-d, -c)$. (5, 0, −1) ga yetish uchun: uchinchi komponent $c = 1$ bo'lishini talab qiladi; keyin $c+$ 2d $= 5d = 2$ beradi; va haqiqatan ham 2c $-d = 0$. Javob: ha,$u+$ 2v $= (5, 0, -1)$.

4. ar $= (2,\ -1), s = (-6,\ 3)$**. Barcha kombinatsiyalarni cr + ds ni tasviring; vektorlar** R^2 **ni to'ldiradimi? Bog'liqlik munosabatini toping.**

[Manba: TA masalalari, 1 hafta, 4-masala]

Yechim. $s = -3$r bo'lgani uchun, har qanday kombinatsiya $(c-$ 3d$)r$ ko'rinishida bo'ladi — bu har doim r ning karrali. Span — bu orqali o'tgan to'g'ri chiziq: vektorlar tekislikni cho'zmaydi. 3r $+s = 0$ munosabati trivialemas, bu bog'liqlikni isbotlaydi (qarang (27)).

5. $v = (1, 0, 1), w = (1, 2, 0)$ **deb oling. cv** $+dw = (x, y, z)$ **yozing;** x, y, z **ni bog'laydigan bitta tenglamani toping;** $b = (4, 6, 1)$ **ni tekshiring.**

[Manba: TA masalalari, 1 hafta, 5-masala]

Yechim. cv $+dw = (c + d,$ 2d, $c)$. O'qib: $z = c$ va $y =$ 2d, demak $x = z + \frac{y}{2}$ — bu tekislikni tasvirlaydigan yagona chiziqli tenglama. uchun: o'ng tomon $= 1 + \frac{6}{2} = 4 = x$, demak b tekislikda joylashgan; $z = 1, y = 6$ dan: $c = 1, d = 3$.

6. c, d **ni shunday topingki,** $c(1,\ 2) + d(3,\ -1) = (7,\ 3)$**; komponent bo'yicha tenglamalarni yozing va yeching.**

[Manba: TA masalalari, 1 hafta, 6-masala]

Yechim. Komponentlar: $c+$ 3d $= 7$ va 2c $-d = 3$. Ikkinchidan $d =$ 2c − 3; almashtiramiz: $c+$6c$-9 = 7$, demak 7c $= 16$, $c = \frac{16}{7}$, $d = 32_7 - 3 = 11_7$. Ikkala komponent ham aniq tekshiriladi.

7. $u = (1, 1, 0), v = (0, 1, 1), w = (1, 2, 1)$**.** w **yangi yo'nalish qo'shadimi?** span**{**u, v, w**}** **ni tasvirlang; nolga teng bo'lgan munosabatni keltiring.**

[Manba: TA masalalari, 1 hafta, 7-masala]

Yechim. $w = u + v$ (komponent bo'yicha tekshirish: $1+0=1, 1+1=2, 0+1=1$), shuning uchun w ortiqcha. $u + v - w = 0$ munosabati trivialemas, demak uchta vektor bog'liq va span — bu faqat u va v (ular karrali emas) cho'zgan tekislik.

8. Ikkita vektor butun R^3 ni cho'zishi mumkinmi? Uchta vektor R^3 ni cho'zishi mumkin va bog'liq bo'lishi mumkinmi? "Yangi yo'nalishlar" orqali tushuntiring.

[Manba: TA masalalari, 1 hafta, 8-masala]

Yechim. Ikkita vektor maksimal tekislikni cho'zadi — har bir vektor maksimal bitta yangi yo'nalish beradi, ikki yo'nalish bilan uch o'lchamli fazoni to'ldirib bo'lmaydi. Uchta bog'liq vektor ham maksimal tekislikni cho'zadi: bog'liqlik bittasining qolganlarining kombinatsiyasi ekanligini va uchinchi yo'nalish qo'shmasligini anglatadi. R^3 ga erishish uchun uchta mustaqil yo'nalish kerak; shuning uchun uchta bog'liq vektor hech qachon R^3 ni cho'zmaydi.

3 2-Hafta — Skalyar va vektor ko'paytmalari, matritsalarga kirish (Lineal algebra I)

3.1 Lineal algebra I, 2-hafta: formulalar katalogi

2-hafta vektor geometriyasining ikki ko'paytmasi va birinchi matritsa tushunchalarini qo'shadi.

$$u \cdot v = u_1 v_1 + \cdots + u_n v_n = u^T v, \quad \cos\theta = \frac{u \cdot v}{\|u\|\,\|v\|} \tag{45}$$

$$u \perp v \Leftrightarrow u \cdot v = 0, \quad \text{proj}_v\, u = \frac{u \cdot v}{v \cdot v}\, v \tag{46}$$

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}, \quad \|u \times v\| = \|u\|\,\|v\|\,|\sin\theta| \text{ (yuza)} \tag{47}$$

$$u \times v = -(v \times u), \quad (u \times v) \perp u, v \tag{48}$$

$$A = [a_{ij}] \in \mathbb{R}^{m \times n}, \quad (A+B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\, a_{ij} \tag{49}$$

$$Ax = x_1 a_1 + \cdots + x_n a_n \text{ (ustunlar rasmi)} \tag{50}$$

bunda a_j — A ning $j-$ustuni. O'lchamlar mos bo'lishi shart: Ax faqat $Am \times n$ va $x \in R^n$ bo'lganda mavjud, natija R^m da yotadi. Diagonal matritsa $D = diag(d_1, \ldots, d_n)$ har bir koordinatani masshtablaydi: $Dx = (d_1 x_1, \ldots, d_n x_n)^T$.

3.2 Lineal algebra I, 2-hafta: ma’ruzadagi auditoriya masalalari — savollar va to’liq yechimlar

[3][]4

1. $u =, v =$**.** $u \cdot v$ **ni hisoblang; burchakni tasniflang.** $-2\ 5$

[Manba: LA ma’ruzasi, 2-hafta, 1-masala]

Yechim. $u \cdot v = 3 \cdot 4 + (-2) \cdot 5 = 12 - 10 = 2 > 0$. Musbat skalyar ko‘paytma $\cos\theta > 0$ degani, ya‘ni burchak o‘tkir.

2. $u = \begin{bmatrix} 2 \\ -1 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 3 \\ 0 \end{bmatrix}$ **bo‘lsin:** $\|u\|$ **va** $dist(u, v)$ **ni toping.**

[Manba: LA ma’ruzasi, 2-hafta, 2-masala]

Yechim. $\|u\| = \sqrt{4+1+4} = 3$. $u - v = \begin{bmatrix} 3 \\ -4 \\ 2 \end{bmatrix}$, demak $dist(u,\, v) = \sqrt{9+16+4} = \sqrt{29}$.

3. $u = \begin{bmatrix} 2 \\ k \\ -1 \end{bmatrix}$ **vektoriga perpendikular bo’ladigan** $v = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix}$ **uchun** k **ni toping.**

[Manba: LA ma’ruzasi, 2-hafta, 3-masala]

Yechim. Perpendikulyarlik — bu $u \cdot v = 0$: $6 + k - 4 = 0$, demak $k = -2$.

4. $u =$ **vav = o‘rtasidagi burchakni toping.** $1\ -1$

[Manba: Lektsiya LA, nedelya 2, zadacha 4]

Yechim. $u \cdot v = 1 - 1 = 0$: vektorlar perpendikulyar, $\theta = 90^\circ$ — formulalar kerak emas, skalyar ko‘paytma nolga teng bo‘lganda.

$$\cos\varphi = (u \cdot v)/(||u||||v||)$$

5. $u = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$ **ni** $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ **ga proyeksiyaling.**

[Manba: Lektsiya LA, nedelya 2, zadacha 5]

Yechim. $u \cdot v = 6, v \cdot v = 2$, shuning uchun $proj_vu = (\frac{6}{2})v = 3\text{v} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$. Qoldiq $u-$ $3\text{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ shartni qanoatlantiradi: $(u-\ 3\text{v}) \cdot v = 0$ — parallel va perpendikular qismlarga ajralish.

6. $u = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix}$, $v = \begin{bmatrix} -1 \\ 2 \\ 0 \end{bmatrix}$ **uchun** $u \times v$ **ni hisoblang va ortogonallikni tekshiring.**

$$2 \cdot 2 - 0 \cdot (-1)[4]$$

[Manba: Lektsiya LA, nedelya 2, zadacha 6]

Yechim. (47) ga ko‘ra:u ×v $= 0\cdot 3 - 1\cdot 2 = -2$. Tekshirish: $1\cdot 4 + 2\cdot(-2) + 0\cdot$

$$u \times v = \begin{bmatrix}2\\0\\2\end{bmatrix} \times \begin{bmatrix}-1\\2\\0\end{bmatrix} = [0\cdot 0 - 2\cdot 2;$$

$$2\cdot(-1) - 2\cdot 0; 2\cdot 2 - 0\cdot(-1)] = \begin{bmatrix}-4\\-2\\4\end{bmatrix} (16)$$

$(-7) = 0$ va $3\cdot 4 + (-1)(-2) + 2\cdot(-7) = 0$ — ikkala kirishga ham ortogonal. 2 1

7. Tomonlari $a = \begin{bmatrix}2\\0\\1\end{bmatrix}$, $b = \begin{bmatrix}1\\3\\-1\end{bmatrix}$ **bo‘lgan parallelogramm yuzini toping.**

$$0\cdot(-1) - 1\cdot 3 - 3$$

[Manba: Lektsiya LA, nedelya 2, zadacha 7]

Yechim. $a \times b = [0\cdot(-1) - 1\cdot 3;\ 1\cdot 1 - 2\cdot(-1);\ 2\cdot 3 - 0\cdot 1] = \begin{bmatrix}-3\\3\\6\end{bmatrix}$, shuning uchun yuza $\|a \times b\| = \sqrt{9} + 9 + 36 = \sqrt{54} = 3\sqrt{6}$.

$$2\cdot 3 - 0\cdot 1 = 6$$

8. $A =$ **uchun o‘lchamni ko‘rsating,** a_{13}, a_{22} **ni toping va ustunlarni yozing** $035a_1, a_3$**.**

[Manba: Lektsiya LA, nedelya 2, zadacha 8]

Yechim. A o‘lchami 2×3 (avval qatorlar). $a_{13} = 4(1-$qator, 3-ustun$), a_{22} = 3.a_1 = , a_3 =$. 0 5

$$A = \begin{bmatrix}2 & 4 & \\ 0 & 3 & 5\end{bmatrix}$$

9. $A = \begin{bmatrix}1 & -2\\3 & 0\end{bmatrix}$, $B = \begin{bmatrix}4 & 1\\-1 & 5\end{bmatrix}$ **bo‘lsin. 2A** $-B$ **ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 9]

Yechim. 2A $= \begin{bmatrix}2 & -4\\6 & 0\end{bmatrix}$; element bo‘yicha ayiramiz: 2A $-B = [2-4, -4-1;\ 6-(-1),$ $0-5] = \begin{bmatrix}-2 & -5\\7 & -5\end{bmatrix}$.

10. A **o‘lchami** $3\times 2, B$ **—** $2\times 3, x \in R^2, y \in R^3$**.** Ax, Ay, Bx**, By dan qaysilari aniqlangan?**

[Manba: Lektsiya LA, nedelya 2, zadacha 10]

Yechim. Ax: $(3\times2)(2\times1)OK$, natija R^3 da. By: $(2\times3)(3\times1)OK$, natija R^2 da. Ay mos kelmaydi $(2 \neq 3)$ va Bx mos kelmaydi $(3 \neq 2)$: ikkalasi ham aniqlanmagan. Faqat ichki o‘lchamlar mos kelishi kerak. 4

11. $D = diag(2,\,-1,\,3)$, $x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$ **bo’lsin.** Dx **va** I**3x ni toping.**

[Manba: Lektsiya LA, nedelya 2, zadacha 11]

Yechim. $Dx = [2\cdot4;\,-1\cdot5;\,3\cdot(-2)] = \begin{bmatrix} 8 \\ -5 \\ -6 \end{bmatrix}$: har bir koordinata o’z diagonal elementi bilan masshtablanadi. I3x $= x = \begin{bmatrix} 4 \\ 5 \\ -2 \end{bmatrix}$: birlik matritsa hech narsa qilmaydi.

$$3\cdot(-2) = -6$$

. $I_3x = x = 5$: birlik matritsa hech narsa qilmaydi. -2

12. $A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 4 & 0 & 8 \end{bmatrix}$, $x = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$ **bo’lsin.** Ax **ni ustunlar tasviri va qatorlar skalyar ko’paytmalari orqali hisoblang.**

$$\begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 0 & 0 & 8 \end{bmatrix}$$

[Manba: Lektsiya LA, nedelya 2, zadacha 12]

Yechim. Ustunlar: $2\cdot\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix} + (-1)\cdot\begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix} + 0\cdot\begin{bmatrix} 0 \\ 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Qatorlar: $\begin{bmatrix} 1 & 2 \end{bmatrix}\cdot x = 0$, $\begin{bmatrix} -1 & 3 \end{bmatrix}\cdot x = -5$, $\begin{bmatrix} 4 & 0 \end{bmatrix}\cdot x = 8$; ustunma-ustun xuddi shu $\begin{bmatrix} 0 \\ -5 \\ 8 \end{bmatrix}$. Ikki tasvir har doim mos keladi.

$$L(x) = Axformulasi\ bilan\ aniqlanadi$$

$$12].$$

13. $A =$ **va** $Ax =$ **berilgan;** x **ni toping. 3 −1 5**

[Manba: Lektsiya LA, nedelya 2, zadacha 13]

Yechim. Komponentlar: $x_1 + 2x_2 = 7$, $3x_1 - x_2 = 5$. Ikkinchisidan $x_2 = 3x_1 - 5$;

$$x = [\frac{17}{7}]$$

almashtiramiz: $x_1 + 6x_1 - 10 = 7$, demak $7x_1 = 17$, $x_1 = \frac{17}{7}$, $x_2 = 16_7$. Shunday qilibx =. 16/7

14. Nol bo‘lmagan vektorlar $u+v$ va $u-v$ ortogonal. $\|u\|$ va $\|v\|$ haqida nima deyish mumkin?

[Manba: Lektsiya LA, nedelya 2, zadacha 14]

Yechim. Ortogonallik $0 = (u+v)\cdot(u-v) = \|u\|^2 - \|v\|^2$ (kesishuvchi hadlar bekor qilinadi), bundan $\|u\| = \|v\|$: teng uzunliklar, vau vav — to‘g‘ri burchakli parallelogrammning yarim diagonallari.

3.3 Chiziqli algebra I, nedelya 2: mashqlar ISA — savollar va to‘liq yechimlar

1. $u = (-3,\ 2)^T, v = (4,\ 5)^T$ bo‘lsin. $u \cdot v$ ni hisoblang; burchak o‘tkir, to‘g‘ri yoki o‘tkirmi?

[Manba: LA ISA, nedelya 2, mshr. 1]

Yechim. $u \cdot v = -12 + 10 = -2 < 0$, demak $\cos\theta < 0$: burchak o‘tkir.

2. aa $= (1,\ -2,\ 4)^T, b = (3,\ 0,\ -1)^T$. Topinga $\cdot b$, $\|a\|$, $\|b\|$, $\|a-b\|$.

[Manba: ISA LA, hafta 2, mashq 2]

Yechim. $a \cdot b = 3 + 0 - 4 = -1$; $\|a\| = \sqrt{1+4+16} = \sqrt{21}$; $\|b\| = \sqrt{9+0+1} = \sqrt{10}$; $a - b = (-2,\ -2,\ 5)^T$, $\|a-b\| = \sqrt{4+4+25} = \sqrt{33}$.

3. $u = (2, k, 3)^T$ vektori $v = (-1, 4, 2)^T$ vektoriga perpendikular bo’ladigan k ni toping.

[Manba: ISA LA, hafta 2, mashq 3]

Yechim. $u \cdot v = -2 + 4\text{k} + 6 = 0 \Rightarrow 4\text{k} = -4 \Rightarrow k = -1$.

4. $u = (3,\ 1)^T$, $v = (-2,\ 4)^T$ uchun aniq $\cos\theta$ ni, keyin θ ni gradus aniqligi bilan toping.

[Manba: ISA LA, hafta 2, mashq 4]

Yechim. $u \cdot v = 3\cdot(-2) + 1\cdot 4 = -6 + 4 = -2$; $\|u\| = \sqrt{10}, \|v\| = \sqrt{20} = 2\sqrt{5}$; $\cos\theta = -2/\sqrt{200} = -2/(10\sqrt{2}) = -\sqrt{2}/10 \approx -0.141$, demak $\theta \approx 98°$.

$$u \cdot v = 3\cdot(-2) + 1\cdot 4 = -6 + 4 = -2; ||u|| = \sqrt{1}0, ||v|| = \sqrt{2}0 = 2\sqrt{5};$$
$$\cos\theta = (u\cdot v)/(||u||\cdot||v||) =$$

$$(-2)/(\sqrt{1}0 \cdot 2\sqrt{5}) = (-1)/(\sqrt{5}0) = (-1)/(5\sqrt{2}) = -\sqrt{\frac{2}{10}}$$

$-\frac{1}{2} \approx -0.1414$, shuning uchun $\theta \approx 98^\circ$. $\sqrt{5}$

5. $u = (6,1)^T$ vektorining $v = (2,-1)^T$ vektoriga proyeksiyasini toping; $u - proj$ ning v ga perpendikularligini tekshiring.

[Manba: ISA LA, hafta 2, mashq 5]

Yechim. $u \cdot v = 12 - 1 = 11, v \cdot v = 5$: $proj_v u = \frac{11}{5}(2,\ -1)^T = (22_5,\ -\frac{11}{5})^T$. Qoldiq: $u - proj = (\frac{8}{5}, 16_5)^T$; v bilan skalyar ko’paytma: $16_5 - \frac{16}{5} = 0$. Tekshirildi.

6. Nolga teng bo’lmagan p, q vektorlari $p \cdot q = \|p\|\|q\|$ shartni qanoatlantiradi. Ular yo’nalishi haqida qanday xulosa chiqarish mumkin?

[Manba: ISA LA, hafta 2, mashq 6]

Yechim. (45) dan, $\cos\theta = 1$, demak $\theta = 0$: vektorlar bir xil yo'nalishda (biri ikkinchisining musbat karrali). Burchak formulasidagi tenglik Koshi-Bunykovskiy-Shvarts g'oyasining chegaraviy holati.

7. $a = (2,\ 1,\ -1)^T, b = (0,\ 3,\ 2)^T$ **uchun a×b va b×a ni hisoblang; belgilar munosabatini tekshiring.**

$$[2 \cdot 3 - (-1) \cdot 0; 5 - 5]$$

[Manba: ISA LA, hafta 2, mashq 7]

Yechim. $a \times b = [1 \cdot 2 - (-1) \cdot 3;\ (-1) \cdot 0 - 2 \cdot 2;\ 2 \cdot 3 - 1 \cdot 0] = \begin{bmatrix} 5 \\ -4 \\ 6 \end{bmatrix}$, va $b \times a = \begin{bmatrix} -5 \\ 4 \\ -6 \end{bmatrix} = -(a \times b)$ — ishora munosabatini tasdiqlaydi.

$$[2 \cdot 3 - 1 \cdot 0; 1 \cdot 0 - (-1) \cdot 3; (-1) \cdot 3 - 2 \cdot 2] = \begin{bmatrix} 6 \\ 3 \\ -7 \end{bmatrix}$$

(48) ga mos keladi.

8. Mashq 7 dan olingan a ×b vektorining a va b ikkalasiga ham perpendikular ekanligini tekshiring.

[Manba: ISA LA, hafta 2, mashq 8]

Yechim. $a \cdot (a \times b) = 2 \cdot 5 + 1 \cdot (-4) + (-1) \cdot 6 = 10 - 4 - 6 = 0; b \cdot (a \times b) = 0 \cdot 5 + 3 \cdot (-4) + 2 \cdot 6 = -12 + 12 = 0$. Ikkalasi ham nolga teng, (47) dan keyin kafolatlanganidek.

9. $u = (1,\ 2,\ 2)^T, v = (3,\ -1,\ 0)^T$ **tomonli parallelogrammning yuzini toping.**

$$[2 \cdot 0 - 2 \cdot (-1); 2 \cdot 3 - 1 \cdot 0; 1 \cdot (-1) - 2 \cdot 3]$$

; yuz $= \sqrt{4} + 36 + 49 = \sqrt{}$

[Manba: ISA LA, hafta 2, mashq 9]

Yechim. $u \times v = [2 \cdot 0 - 2 \cdot (-1);\ 2 \cdot 3 - 1 \cdot 0;\ 1 \cdot (-1) - 2 \cdot 3] = \begin{bmatrix} 2 \\ 6 \\ -7 \end{bmatrix}$, norma $= \sqrt{4} + 36 + 49 = \sqrt{89}$.

$$||u \times v|| = \sqrt{(2 \cdot 0 - 2 \cdot (-1))^2 + (2 \cdot 3 - 1 \cdot 0)^2 + (1 \cdot (-1) - 2 \cdot 3)^2} = \sqrt{4 + 36 + 49}$$

$$\sqrt{8}9$$

10. $P = (1,\ 0,\ 1)$, $Q = (3,\ 1,\ 0)$, $R = (0,\ 2,\ 4)$**:** PQR **uchburchakning yuzini toping.**

[Manba: ISA LA, hafta 2, mashq 10]

Yechim. P dan chiqadigan tomonlar: $PQ = (2,\ 1,\ -1)^T$, $PR = (-1,\ 2,\ 3)^T$. Vektor ko'paytmasi: $PQ \times PR = (1 \cdot 3 - (-1) \cdot 2; (-1) \cdot (-1) - 2 \cdot 3; 2 \cdot 2 - 1 \cdot (-1)) = (5,\ -5,\ 5)$, norma

$$\sqrt{25 + 25 + 25} = \sqrt{75} = 5\sqrt{3}$$

; uchburchak yuzi $= (\frac{1}{2}) \cdot 5\sqrt{3} = 5\sqrt{3}/2$ (parallelogramm yuzining yarmi). u va v ni almashtirish vektor ko'paytmasining ishorasini teskarisiga o'zgartiradi, yuz o'zgarmaydi.

11. $a = (1, t, 2)^T$ **va** $b = (3, 6, 6)^T$ **vektorlari** a **×b** $= 0$ **shartni qanoatlantiradigan barcha haqiqiy** t **larni toping; agar bunday** t **lar bo'lmasa, tushuntiring.**

[Manba: ISA LA, hafta 2, mashq 11]

Yechim. Parallel vektorlar nol vektor ko'paytmasini beradi, demak $a = $ cb bo'lishi kerak: $1 = 3\text{c} \Rightarrow c = \frac{1}{3}$; unda $t = 6\text{c} = 2$; uchinchisini tekshiramiz: $2 = 6\text{c} = 2$ — mos keladi. Shunday qilib $t = 2$ yagona qiymat (unda $a = \frac{1}{3}b$).

12. $a = (1,\ 0,\ 2)^T, b = (2,\ 1,\ -1)^T, c = (0,\ 3,\ 1)^T$**.** $a \cdot (b$ **×c) ni hisoblang; nol qiymatini talqin qiling.**

$$1 \cdot 1 - (-1) \cdot 34$$

[Manba: ISA LA, hafta 2, mashq 12]

Yechim. $b \times c = [1 \cdot 1 - (-1) \cdot 3;\ (-1) \cdot 0 - 2 \cdot 1;\ 2 \cdot 3 - 1 \cdot 0] = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}$; $a \cdot (b \times c) = 1 \cdot 4 + 0 \cdot (-2) + 2 \cdot 6 = 16$.

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & -1 \\ 0 & 3 & 1 \end{bmatrix}$$

Bu uch vektorga tortilgan parallelepipedning belgilangan hajmi; nol qiymati komplanarlikni (3D hajmi yo'qligini) anglatardi. Bu yerda $|16| = 16$ kub birlik. 2 −1 0

13. $A = \begin{bmatrix} 2 & -1 & 0 \\ 4 & 3 & 5 \end{bmatrix}$ **matritsasi berilgan. O'lchamini ko'rsating;** a_{12}, a_{23} **elementlarini toping; har bir qator va ustunni vektor sifatida yozing.**

[Manba: ISA LA, hafta 2, mashq 13]

Yechim. O'lchami 2×3 (avval qatorlar). $a_{12} = -1 (1-$qator, 2-ustun$), a_{23} = 5$. Qatorlar: $\begin{bmatrix} 2 & -1 & 0 \end{bmatrix}, \begin{bmatrix} 4 & 3 & 5 \end{bmatrix}$; ustunlar: $\begin{bmatrix} 2 \\ 4 \end{bmatrix}, \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 0 \\ 5 \end{bmatrix}$.

$$[1][]35 - 1$$

14. $A = \begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}, B = \begin{bmatrix} 5 & -1 \\ 0 & 2 \end{bmatrix}$ **matritsalar berilgan.** $A + B, A - B, -$**2A, 3A − 2B ni hisoblang.**

[Manba: ISA LA, hafta 2, mashq 14]

Yechim. $A + B = \begin{bmatrix} 6 & 2 \\ -2 & 6 \end{bmatrix}$; $A - B = \begin{bmatrix} -4 & 4 \\ -2 & 2 \end{bmatrix}$; −2A $= \begin{bmatrix} -2 & -6 \\ 4 & -8 \end{bmatrix}$; 3A − 2B $= [3 - 10, 9 + 2;\ -6 - 0,\ 12 - 4] = \begin{bmatrix} -7 & 11 \\ -6 & 8 \end{bmatrix}$.

15. A **matritsasi** 4×3 **o'lchamda,** B **—** $3\times 2, x\in R^2, y\in R^3, z\in R^4$**.** Ax, Ay, Az, Bx**, By,** Bz **dan qaysilari aniqlangan va ularning chiqish o'lchamlari qanday?**

[Manba: ISA LA, hafta 2, mashq 15]

Yechim. Ay: $(4\times 3)(3\times 1)$ — aniqlangan, natija R^4. Bx: $(3\times 2)(2\times 1)$ — aniqlangan, natija R^3. Qolganlari (Ax, Az, By, Bz) ichki o'lchamlari mos kelmaydi va aniqlanmagan.

16. $D = diag(-2,\ 3,\ 1), x = (4,\ -1,\ 5)^T$ **berilgan.** Dx **ni hisoblang;** D **har bir koordinataga nima qilishini tushuntiring.**

$$Dx = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, x = (2, -1, 3)^T.$$

[Manba: ISA LA, hafta 2, mashq 16]

Yechim. Dx: birinchi koordinata -2 ga masshtablanadi (aks etish bilan cho'zish), ikkinchi — 3 ga, uchinchi — 1/2 ga.

17. $A =$ **matritsasi berilgan.** Ax **ni** (a) **ustunlarning 4 0 −1 kombinatsiyasi sifatida,** (b) **qatorlarning skalyar ko'paytmalari orqali hisoblang.**

$$1 - 22 + 2 + 9 = 13$$

[Manba: ISA LA, hafta 2, mashq 17]

Yechim. $(a) 2 - 1 + 3 =$

$$2\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}2\\4\\6\end{bmatrix} - \begin{bmatrix}1\\2\\0\end{bmatrix} + \begin{bmatrix}3\\0\\3\end{bmatrix}$$

$$\begin{bmatrix}4\\2\\9\end{bmatrix}$$

. $(b) 1-$qator: $1\cdot 2 + (-2)(-1) + 3\cdot 3 = 13$; 2-qator: $4\cdot 2 + 0\cdot(-1) + (-1)\cdot 3 = 5$; 3-qator: $2\cdot 2 + 2\cdot(-1) + 2\cdot 3 = 6 - 2 + 6 = 10$.

$$\begin{bmatrix}1 & 4 & 2\\-2 & 0 & 2\\-1 & -1 & 2\end{bmatrix}\begin{bmatrix}2\\-1\\3\end{bmatrix} = \begin{bmatrix}13\\5\\10\end{bmatrix}$$

Xuddi shu natija, bo'lishi kerakgidek.

18. $a_1 = (1,\ 2)^T, a_2 = (3,\ -1)^T$ **vektorlar berilgan.** $x_1a_1 + x_2a_2 = (11,\ 1)^T$ **tenglamani qanoatlantiruvchi** $x = (x_1,\ x_2)^T$ **ni toping.**

[Manba: ISA LA, hafta 2, mashq 18]

Yechim. Komponentlar: $x_1 + 3x_2 = 11$, $2x_1 - x_2 = 1$. Ikkinchisidan $x_2 = 2x_1 - 1$; keyin $x_1 + 6x_1 - 3 = 11 \Rightarrow 7x_1 = 14 \Rightarrow x_1 = 2$, $x_2 = 3$. 2 4

19. $A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix}$ **matritsasi berilgan.** $Ax = 0$ **tenglamani qanoatlantiruvchi barcha** x **ni tavsiflang; nima uchun ustunlar chiziqli bogliq?**

[Manba: ISA LA, hafta 2, mashq 19]

Yechim. $2x_1 + 4x_2 = 0$ va $-x_1 - 2x_2 = 0$ — bir xil tenglama: $2x_1 = -4x_2$. Yechimlar: $x = t \cdot \begin{bmatrix} -2 \\ 1 \end{bmatrix}$, $t \in R$ — butun yechimlar to'g'ri chiziq. Trivial bo'lmagan kombinatsiya $-2 \cdot a_1 + 1 \cdot a_2 = 0$ mavjud, bu ustunlarning chiziqli bog'liqligidir (2-ustun 1-ustunning -2 barobariga teng).

20. A **matritsasi** 3×2 **o'lchamda,** $A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix}$ **va** $A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix}$**.** A **ni yozing va** $A \begin{bmatrix} -2 \\ 5 \end{bmatrix}$ **ni qator skalyar ko'paytmalarisiz hisoblang.**

[Manba: ISA LA, hafta 2, mashq 20]

Yechim. Ikkita tasvir — bu A ning ustunlari: $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \\ 4 & 0 \end{bmatrix}$. Chiziqlilik: $A \cdot \begin{bmatrix} -2 \\ 5 \end{bmatrix} =$

$$-2Ae_1 + 5Ae_2 = -2 \cdot \begin{bmatrix} 2 \\ -1 \\ 4 \end{bmatrix} + 5 \cdot \begin{bmatrix} 3 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 11 \\ 12 \\ -8 \end{bmatrix}.$$

$-2Ae_1 + 5Ae_2 = -2 - 1 + 52 = 12$. 4 0 −8

21. Bir o'lchamli vektorlar uchun $(u+v) \cdot (u-v)$ **ni oching;** $u + v \perp u - v$ **bo'lishi uchun zarur va yetarli shart** $\|u\| = \|v\|$ **ekanligini isbotlang.**

[Manba: ISA LA, hafta 2, mashq 21]

Yechim. Taqsimlab: $(u+v) \cdot (u-v) = u \cdot u - u \cdot v + v \cdot u - v \cdot v = \|u\|^2 - \|v\|^2$ (o'rtadagi hadlar bekor qilinadi). Perpendikulyarlik bu nolga teng bo'lishini anglatadi, ya'ni $\|u\|^2 = \|v\|^2$, va normalar manfiy bo'lmagani uchun $\|u\| = \|v\|$.

22. O'z-o'zini tekshirish: (1) skalyar ko'paytma va vektor ko'paytma qaysi ob'ektni beradi? $(2) u \cdot v$ **belgisi burchakni qanday tasniflaydi? (3) Nima uchun** 2×3 **matritsasi** R^3 **dan vektorni ko'paytiradi, lekin** R^2 **dan emas?** $(4) Ax$ **da** x **komponentlari qanday ishlaydi? (5) 2-haftadagi qaysi amallar teng o'lchamli vektlarni talab qiladi, qaysilari aynan** R^3 **dan vektlarni?**

[Manba: LA ISA, 2-hafta, o'z-o'zini tekshirish]

Yechim. (1) Skalyar; vektor (R^3 da). (2) Musbat ⇒ o'tkir, nol ⇒ to'g'ri, manfiy → o'tmas. $(3) Ax$ ko'paytmasi ichki o'lchamlarning mosligini talab qiladi: 2×3 va 3×1 ishlaydi; 2×1 esa yo'q. (4) Chiziqiy kombinatsiyadagi og'irliklar kabi $x_1 a_1 + \cdots + x_n a_n$. (5) Skalyar ko'paytma, qo'shish va ayirish teng o'lchamli vektorlarni talab qiladi; vektor ko'paytmasi va uning geometriyasi (yuzalar, normalar) faqat R^3 da mavjud.

3.4 Chiziqli algebra I, 2-hafta: TA sessiyasi (15 masala) — savollar va to'liq yechimlar

$$[][] - 25$$

1.$a =, b =$**: hisoblanga** $\cdot b$**; burchakni tasniflang. 4 1**

[Manba: LA TA, 2-hafta, masala 1]

Yechim. $a \cdot b = (-2) \cdot 5 + 4 \cdot 1 = -10 + 4 = -6 < 0$: o'tmas (botiq).

2. $v = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ **vektorining normasini va birlik vektorni toping.**

[Manba: LA TA, 2-hafta, masala 2]

Yechim. $\|v\| = \sqrt{4} + 1 + 4 = 3$; birlik vektor $\hat{v} = [-\frac{2}{3}$; 1/3; 2/3].

3. $p = \begin{bmatrix} -2 \\ 4 \\ 3 \end{bmatrix}$ **va** $q = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ **orasidagi masofani toping.**

[Manba: LA TA, 2-hafta, masala 3]

Yechim. $p - q = \begin{bmatrix} -2 & 2 \\ 4 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} -4 \\ 3 \\ 3 \end{bmatrix}$, masofa $= \sqrt{16} + 9 + 9 = \sqrt{34}$.

4. Agar $\begin{bmatrix} 2 \\ t \\ 4 \end{bmatrix}$ **vektori** $\begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ **ga perpendikulyar bo'lsa,** t **ni toping.**

[Manba: LA TA, 2-hafta, masala 4]

Yechim. $2 - 3t + 8 = 0 \Rightarrow t = \frac{10}{3}$.

$$\begin{bmatrix} 2 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} t \\ -3 \end{bmatrix} = 0 \Rightarrow 2t + 3 = 0 \Rightarrow$$

$$t = -\frac{3}{2}$$

5. $u =, v =$ **uchun** $\cos\theta$ **ni toping. 1 3**

$$\cos\theta = 1/\sqrt{50} = \sqrt{\frac{2}{10}}$$

[Manba: LA TA, 2-hafta, masala 5]

Yechim. $u \cdot v = 2 \cdot (-1) + 1 \cdot 3 = 1$; $\|u\| = \sqrt{5}, \|v\| = \sqrt{10}$; $\cos\theta = 1/\sqrt{50} = \sqrt{2}/10 \approx 0.141$, $\theta \approx 82°$.

6. $u = (5, -1)^T$ **ni** $v = (2, 1)^T$ **ga proyeksiya qiling;** u **ni parallel + perpendikulyar qismlarga yozing.**

$$[\frac{7}{5}; \frac{18}{5}]$$

[Manba: LA TA, 2-hafta, masala 7]

Yechim. $u \cdot v = 9, v \cdot v = 5$: $proj =$

$$(\frac{9}{5})\begin{bmatrix}2\\1\end{bmatrix}$$

. Perpendikulyar qism:

$$\begin{bmatrix}5\\-1\end{bmatrix}-[\frac{14}{5};\frac{9}{5}]$$

$u - proj =$

$$[\frac{11}{5};-\frac{18}{5}]$$

. Yoyilma:

$$u=[\frac{14}{5};\frac{9}{5}]+[\frac{11}{5};-\frac{18}{5}]$$

.

7. $a=\begin{bmatrix}2\\-1\\3\end{bmatrix}$, $b=\begin{bmatrix}0\\4\\1\end{bmatrix}$ **uchun** $a\times b$ **ni hisoblang.**

$$(-1)\cdot 1-3\cdot 4-13$$

[Manba: LA TA, 2-hafta, masala 8]

Yechim. $a\times b=[(-1)\cdot 1-3\cdot 4;\ 3\cdot 0-2\cdot 1;\ 2\cdot 4-(-1)\cdot 0]=\begin{bmatrix}-13\\-2\\8\end{bmatrix}$.

$$|-48;0-1|$$

8. $A=(0,\,0,\,0)$, $B=(2,\,1,\,0)$, $C=(1,\,0,\,3)$**: uchburchak yuzasini toping.**

$$\begin{bmatrix}2&1&0\\1&0&3\end{bmatrix}$$

[Manba: LA TA, 2-hafta, masala 9]

Yechim. $AB=AC=$ vektor ko'paytmasi

$$\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}^{1/2}$$

$=0\cdot 1-2\cdot 3=-6$, norma $=46$; yuzasi $=\frac{46}{2}$.

$$S=(\frac{1}{2})\cdot||(2;1;0)\times(1;0;3)||=(\frac{1}{2})\cdot||(3;-6;$$
$$-1)||=(\frac{1}{2})\cdot\sqrt{3^2+(-6)^2+(-1)^2}=(\frac{1}{2})\cdot\sqrt{46}$$

9. $M =$ **uchun o’lcham,** m_{21}**, barcha ustunlar. 4 3 5**

$$\begin{bmatrix} 1 & 0 & -2 \\ 4 & 3 & 5 \end{bmatrix}$$

[Manba: LA TA, 2-hafta, masala 10]

Yechim. 2×3; $m_{21} = 4$; ustunlar„. 4 3 5

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 4 & 3 \\ 5 & 0 & 0 \end{bmatrix}$$

10. $A = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 4 \\ 2 & -2 \end{bmatrix}$**. 3A + 2B ni toping.**

[Manba: LA TA, 2-hafta, masala 11]

Yechim. 3A $= \begin{bmatrix} 6 & 3 \\ -3 & 9 \end{bmatrix}$, 2B $= \begin{bmatrix} 0 & 8 \\ 4 & -4 \end{bmatrix}$; yig’indi $= \begin{bmatrix} 6 & 11 \\ 1 & 5 \end{bmatrix}$.

11. $A = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ -2 \\ 3 \end{bmatrix}$**.** Ax **ni ustunlar kombinatsiyasi sifatida hisoblang.**

[Manba: LA TA, 2-hafta, masala 12]

Yechim. $1 - 2 + 3 ==$.

$$\begin{bmatrix} 0 & 4 & 1 \\ -2 & 0 & 4 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix} + y \begin{bmatrix} 4 \\ 0 \\ 3 \end{bmatrix} + z \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = [0x + 4y + 1z;$$
$$-2x + 0y + 4z;$$

$$1x + 3y + 2z] = \begin{bmatrix} 4y & + & z & \\ -2x & + & 4z & \\ x & + & 3y & + & 2z \end{bmatrix} = \begin{bmatrix} 9 \\ 5 \\ 13 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 2 \\ 2 \end{bmatrix} + \begin{bmatrix} 9 \\ 13 \end{bmatrix}$$

12. $a_1 =, a_2 =, b =$**.** $[a_1 a_2]x = b$ **bo’ladigan barcha** x **larni toping. 2 4 6**

[Manba: LA TA, 2-hafta, masala 13]

Yechim. Ikkala tenglama ham $x_1 + 2x_2 = 3$ ga kelib chiqadi (tizma mos, lekin

$$\begin{bmatrix} 3 & & 2t \\ -7 & 2 & \\ -9 & 4 & \end{bmatrix}$$

ortiqcha, chunki $a_2 = 2a_1$ va $b = 3a_1$). Umumiy yechim:$x =, t \in R$ — t bir xil kombinatsiya uchun cheksiz ko'p og'irliklar.

13. Koordinatalarni ochmasdan, nima uchun $(u \times v) \cdot (u + v) = 0$ ekanini tushuntiring.

[Manba: LA TA, 2-hafta, masala 14]

Yechim. Vektor ko'paytmasi ikkala kirish vektoriga ham perpendikulyar, shuning uchun ularning har qanday kombinatsiyasiga perpendikulyar: $(u \times v) \cdot u = 0$ va $(u \times v) \cdot v = 0$; qo'shib, $u + v$ bilan skalyar ko'paytma $0 + 0 = 0$ ga teng.

$$[u_1 u_2 u_3] \times [v_1 v_2 v_3] \cdot ([u_1 u_2 u_3] + [v_1 v_2 v_3]) = 0$$

14. A o'lchami 2×2, bunda $A \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$ va $A \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. A va $A \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix}$ ni toping.

[Manba: TA LA, 2-hafta, masala 15]

Yechim. Standart bazis vektorlarining tasvirlari — ustunlar:

$$A = \begin{bmatrix} 2 & 3 \\ -1 & 4 \end{bmatrix}$$

. U holda $A \cdot \begin{bmatrix} 5 \\ -2 \end{bmatrix} = 5 \cdot \begin{bmatrix} 2 \\ -1 \end{bmatrix} - 2 \cdot \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 10 & 6 \\ -5 & 8 \end{bmatrix} = \begin{bmatrix} 4 \\ -13 \end{bmatrix}$; chiziqlilik hisobni skalyar ko'paytmasiz ikki masshtablashga aylantiradi.

4 3-hafta — Matritsa algebrasi, determinantlar, aralash ko'paytma (Chiziqli algebra I)

4.1 Chiziqli algebra I: 3-hafta formulalari katalogi

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad (cA)_{ij} = c\,a_{ij}, \quad (A^T)_{ij} = a_{ji}, \quad (A^T)^T = A \tag{65}$$

$$(A + B)^T = A^T + B^T, \quad (cA)^T = cA^T, \quad A^T = A \text{ (simmetrik)} \tag{66}$$

$$A \in \mathbb{R}^{m \times n},\ B \in \mathbb{R}^{n \times p} \Rightarrow AB \in \mathbb{R}^{m \times p} \tag{67}$$

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj} \tag{68}$$

$$A(BC) = (AB)C, \quad A(B + C) = AB + AC, \quad (A + B)C = AC + BC \tag{69}$$

$$(AB)^T = B^T A^T, \quad I_m A = A = A I_n \tag{70}$$

$$A\,[\,b_1 \cdots b_p\,] = [\,Ab_1 \cdots Ab_p\,] \text{ (ustunlar qoidasi)} \tag{71}$$

$$u^T v = \text{skalyar}, \quad uv^T = \text{matritsa (tashqi ko'paytma)} \tag{72}$$

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \text{ (ishorali yuza)} \tag{73}$$

$$C_{ij} = (-1)^{i+j} \det M_{ij} \text{ (kofaktor)}, \quad \det A = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \tag{74}$$

bunda Mij — minori (i-qator va $j-$ustun o'chirilgan). Har qanday qator yoki ustun bo'ylab yoyish mumkin; nollari bor qator yoki ustunlar hisobni kamaytiradi. Uchburchak matritsalarda determinant diagonal elementlar ko'paytmasi. Boshlang'ich qator amallari qoidalari 1-jadvalda jamlangan.

A ustidagi amal	**det A ga ta'siri**
Ikkala qatorni almashtirish	ishorasini almashtiradi
Bitta qatorni c ga ko'paytirish	determinantni c ga ko'paytiradi
Bitta qatorning ko'paytmasini boshqasiga qo'shish	o'zgarish yo'q
Nol qator yoki ikki proporsional qator	determinant 0
Transponirlash	$\det(A^T) = \det A$
Bir o'lchamli kvadrat matritsalar ko'paytmasi	$\det(AB) = \det A \det B$

1-jadval. Kvadrat matritsalar uchun determinant qoidalari (qator versiyalari; ustun versiyalari ham o'rinli).

Butun $n \times n$ matritsani masshtablash n ta qatorning hammasini masshtablaydi: det(cA) = c^{n} det A. Determinantlar qo'shishga taqsimlanmaydi: $A = B = I_2$ uchun $\det(A + B) = 4$, holbuki $\det A + \det B = 2$.

$$(u \times v) \cdot w = u \cdot (v \times w) = \det[\, u \; v \; w \,] \text{ (A.K)} \tag{75}$$

$$V_{\text{parallelepiped}} = |(u \times v) \cdot w|, \quad h = \frac{|(u \times v) \cdot w|}{\|u \times v\|} \tag{76}$$

$$V_{\text{tetraedr}} = \frac{1}{6} |\det[\, u \; v \; w \,]|, \quad T = 0 \Leftrightarrow u, v, w \text{ komplanar} \tag{77}$$

4.2 Chiziqli algebra I: 3-hafta ma'ruzasi masalalari (15) — savollar va to'liq yechimlar

$$[][2]1 - 21$$

1. $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -1 & 4 \end{bmatrix}$ **bo'lsin. 2A** $-B$ **ni hisoblang va (2, 1) elementini ko'rsating;** A **ni** 2×3 **matritsasiga qo'shish mumkinmi?**

[Manba: LA ma'ruzasi, 3-hafta, masala 1]

Yechim. $2\text{A} = \begin{bmatrix} 2 & -4 \\ 6 & 0 \end{bmatrix}$; $2\text{A} - B = [2-2, -4-1;\ 6-(-1),\ 0-4] = \begin{bmatrix} 0 & -5 \\ 7 & -4 \end{bmatrix}$; (2, 1) elementi $2(3) - (-1) = 7$. 2×3 matritsasi bilan qo'shish aniqlanmagan — ikkala o'lcham ham mos kelishi kerak, umumiy elementlar soni emas.

2. $C = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 2 \end{bmatrix}$ **bo'lsin.** C^T **ni uning o'lchami bilan yozing;** CC^T **ni hisoblang;** u **simmetrikmi?**

[Manba: LA ma'ruzasi, 3-hafta, masala 2]

Yechim. $C^T = \begin{bmatrix} 1 & 0 \\ 2 & 3 \\ 1 & 2 \end{bmatrix}$ (3×2). $CC^T = [1+4+1, 0+6-2;\ 0+6-2,\ 0+9+4] = \begin{bmatrix} 6 & 4 \\ 4 & 13 \end{bmatrix}$. Diagonall elementlar — o'zlari bilan skalyar ko'paytirilgan qatorlar; tashqi diagonallari teng — matritsa simmetrik.

3. Yuqoridagi C **va** $x = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$ **uchun** Cx **ni ikki usul bilan hisoblang.**

[Manba: LA ma'ruzasi, 3-hafta, masala 3]

Yechim. Qatorlar: $1 \cdot 2 + 2 \cdot (-1) + (-1) \cdot 1 = -1$; $0 + 3 \cdot (-1) + 2 \cdot 1 = -1$; demak $Cx = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. Ustunlar: $2 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + (-1) \cdot \begin{bmatrix} 2 \\ 3 \end{bmatrix} + 1 \cdot \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$. O'lchamlar $(2 \times 3)(3 \times 1) 2 \times 1$ chiqish beradi.

Ustunlar: $2 - + =$. O'lchamlar $(2 \times 3)(3 \times 1)$ chiqish 2×1 beradi.

$$C = \begin{bmatrix} 0 & 3 & 2 \\ -1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix}$$

4. 1-masaladagi A, B **uchun** AB **va** BA **ni hisoblang. Tenglar?**

$$\begin{bmatrix} 5 & 1 & 2 \\ -2 & -1 & 1 \\ 1 & -2 & 4 \end{bmatrix}$$

[Manba: LA ma'ruzasi, 3-hafta, masala 4]

Yechim. $AB =$

$$\begin{bmatrix} 6 & 3 \\ 8 & 7 \end{bmatrix}$$

, $BA =$

$$\begin{bmatrix} 8 & 10 \\ 0 & 0 \end{bmatrix}$$

. Turlicha; hatto ikkala tartibda ham mavjud bo'lganda, tenglik kafolatlanmaydi.

$$\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix},$$

$$D = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

5. $C = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 2 \end{bmatrix}$, $D = \begin{bmatrix} 2 & 0 \\ 1 & -1 \\ 3 & 2 \end{bmatrix}$ **bo'lsin.** CD **va** DC **o'lchamlarini oldindan aytib, ikkalasini hisoblang.**

[Manba: LA ma'ruzasi, 3-hafta, masala 5]

Yechim. CD: $(2\times 3)(3\times 2) = 2\times 2$; DC: $(3\times 2)(2\times 3) = 3\times 3$. $CD = [2+2-3, 0-2-2;$ $0+3+6, 0-3+4] = \begin{bmatrix} 1 & -4 \\ 9 & 1 \end{bmatrix}$; $DC = \begin{bmatrix} 2 & 4 & -2 \\ 1 & -1 & -3 \\ 3 & 12 & 1 \end{bmatrix}$.

6. $S =$ **(siljish)**, $H =$ **(cho'zish)**, $x =$**. HSx va**

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

SHx ni hisoblang; har bir aylantirishni tavsiflang.

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 6]

Yechim. $Sx =$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

, keyin HSx =

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

; shuningdek $Hx =$

$$\begin{bmatrix} 2 & 4 \\ 2 & 3 \end{bmatrix}$$

, keyin SHx =

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & \end{bmatrix}$$

. HS da siljish hissasi keyin cho'ziladi; SH da esa avval cho'zish, keyin ustiga siljish qo'shiladi. Ikkala operatsiya bir xil, bosh tartibda, bosh natija: matritsalar almashmaydi.

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

,$v =$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

: hisoblangu$^T v$ vauv^T; tekshiring (uv$^T)^T$ =vu^T.

7. $u = 20 - 13$ **uchun**

[Manba: LA darsi, 3-hafta, masala 7]

Yechim. Ichki: $u^T v = 2 + 0 - 3 = -1$ (skalyar). Tashqi: uv$^T =$

$$\begin{bmatrix} 2 & 0 & -1 \\ 0 & 0 & 0 \\ 3 & 0 & -3 \end{bmatrix}$$

(3×3). Element bo'yicha transponirlab, (uv$^T)^T =$ (vu$^T)_{ij}$, shuning uchun (uv$^T)^T =$ vu^T. Yozilgan matritsalar — 3×3 o'zaro transponirlangan.

$$(AB)^T = B^T A^T$$

.

8. Masala 1 dan A, B **dan foydalanib,** $A^T B^T$ **mustaqil hisoblang; mos keladimi?**

$$[4]4 - 76$$

[Manba: LA darsi, 3-hafta, masala 8]

Yechim. $AB =$

$$\begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

, demak $(AB)^T =$

$$\begin{bmatrix} 3 & -10 \\ 12 & -14 \end{bmatrix}$$

. Avval

$$\begin{bmatrix} 6 & 3 \\ -7 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 12 \\ -10 & -14 \end{bmatrix}$$

transponirlangan: $A^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}$$

, $B^T =$

$$\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix}$$

; keyin $A^T B^T =$

$$\begin{bmatrix} -2 & 0 \\ 1 & 4 \end{bmatrix}\begin{bmatrix} 4 & -7 \\ 6 & 3 \end{bmatrix} = \begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

. Hisoblaymiz: $(2\cdot 1 + (-1)(-2)) = 4$, $(2\cdot 3 + 0) = 6$

$$\begin{bmatrix} 4 & 6 \\ 28 & -5 \end{bmatrix}$$

$= (AB)^T$, bu (70) ni tasdiqlaydi. Lekin $A^T B^T =$

$$\begin{bmatrix} -8 & -14 \\ 28 & -5 \end{bmatrix}$$

farq qiladi: to'g'ri identifikator ko'paytuvchilar tartibini o'zgartiradi.

9. $E = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ **deb belgilaymiz.** $\det E$ **ni hisoblang; uning ustunlariga tortilgan parallelogramm va uchburchkning maydonlarini toping; orientatsiya saqlanadimi?**

[Manba: LA darsi, 3-hafta, masala 9]

Yechim. $\det E = 3\cdot 2 - 1\cdot 1 = 5$. Parallelogramm maydoni $= |5| = 5$; uchburchk maydoni $= \frac{5}{2}$. Orientatsiya saqlanadi: $\det E > 0$.

10. $F = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & 0 & 2 \end{bmatrix}$ **uchun** $\det F$ **ni birinchi qator kofaktorlari bo'yicha hisoblang; birinchi ustun bo'yicha tekshiring.**

$$\begin{bmatrix} 3 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 3 & 1 \end{bmatrix}$$

[Manba: LA darsi, 3-hafta, masala 10]

Yechim. Birinchi qator: $\det F = 2\det -1\det +0 = 2(6) - 1(-1) = 13.$ 0 2 1 2

$$\det F = 3\det\begin{bmatrix}0 & 2\\ 1 & 0\end{bmatrix} - 1\det\begin{bmatrix}1 & 2\\ 0 & 0\end{bmatrix} + 0\det\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = 3(0) - 1(0) + 0(1)$$

$$0.(3.1)$$

Birinchi ustun: $2\det +1\det = 12 + 1 = 13$ (kofaktor (3, 1) belgisi $0231(-1)^{3+1} = +$). Har qanday qator yoki ustun bir xil qiymat beradi.

11. 3×3 **matritsa** M **uchun** $\det M = 5$ **ma'lum.** N **ni hosil qiling: 1,2-qatorlarni almashtiring; 3-qatorni 2 ga ko'paytiring;** $3\times$**(yangi 1-qator) ni 2-qatorga qo'shing.** $\det N, \det(\mathbf{2M}), \det(M^TM)$ **ni toping.**

[Manba: LA darsi, 3-hafta, masala 11]

Yechim. Almashtirish: belgi o'zgaradi (-5); 3-qatorni masshtablash: yana ko'paytuvchi 2; qator qo'shish: o'zgarishsiz. $\det N = -2 \cdot 5 = -10$. Barcha qatorlarni masshtablash: $\det(2\mathrm{M}) = 2^3 \cdot 5 = 40$. Ko'paytuvchilar qoidasi: $\det(M^TM) = \det M^T \det M = 25$.

$$\det(M^TM) = \det M^T \det M = 25.$$

$= 0, v = 2, w = 1$. Hisoblang $(u \times \mathrm{v}) \cdot w$, hajmlar

12. $u103$ **paralleliped va tetraedr hajmlari; agaru,**v **ni almashtirsak nima o'zgaradi?**

$$[2 \cdot 2 - 0 \cdot 14]$$

[Manba: LA darsi, 3-hafta, masala 12]

Yechim. $u \times v = [1 \cdot 1 - 2 \cdot 0;\ 2 \cdot 0 - 1 \cdot 3;\ 1 \cdot 3 - 2 \cdot 0] = \begin{bmatrix}1\\ -3\\ 3\end{bmatrix}$; w bilan skalyar ko'paytma: $1 \cdot 2 + (-3) \cdot 1 + 3 \cdot 4 = 2 - 3 + 12 = 11$. Hajm

$$2 \cdot 2 - 0 \cdot 14$$

paralleliped 11; tetraedr 11/6. u va v ni almashtirish belgini -11 ga o'zgartiradi, lekin ikkala hajm 11 va 11/6 (absolyut qiymatlar) bo'lib qoladi.

$$0 \cdot 0 - 1 \cdot 2 = -2$$

.

13. $u = \begin{bmatrix}1\\0\\1\end{bmatrix}$, $v = \begin{bmatrix}0\\2\\1\end{bmatrix}$, $w = \begin{bmatrix}1\\1\\k\end{bmatrix}$ **uchun vektorlar koplanar bo'lgan barcha** k **ni toping; shu** k **da** w **ni** u, v **orqali ifodalang.**

$$0 \cdot 1 - 1 \cdot 2 - 2$$

$\Leftarrow\Rightarrow T = 0$**. Bu yerdau** $\times\mathbf{v} = 1 \cdot 0 - 1 \cdot 1 = -1$**,**

[Manba: LA darsi, 3-hafta, masala 13]

Yechim. Komplanarlik

$$1 \cdot 2 - 0 \cdot 02$$

$T = -2 - 1 +$ 2k = 2k $-3 = 0 \Rightarrow k = \frac{3}{2}$ degani. U holdaw $= \alpha u + \beta v$: birinchi ikki koordinata $\alpha = 1, 2\beta = 1 \Rightarrow \beta = \frac{1}{2}$ beradi; uchinchisini tekshiramiz: $1 + 1_2 = 3_2$ — mos keladi,$w = u + \frac{1}{2}v$.

$$W = U + (\frac{5}{7})V,$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} (16)$$

14. $D = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$, $X = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ **bo'lsin.** $DX = XD$ **bo'ladigan barcha** X **ni toping; agar** D **ning ikkala diagonal elementi teng bo'lsa nima bo'ladi?**

[Manba: LA darsi, 3-hafta, masala 14]

Yechim. $DX = \begin{bmatrix} 2a & 2b \\ 3c & 3d \end{bmatrix}$, $XD = \begin{bmatrix} 2a & 3b \\ 2c & 3d \end{bmatrix}$. Tenglik $b = c = 0$ talab qiladi, bunda a va d ixtiyoriy: aynan diagonal matritsalar bu D bilan kommutativ. Agar $D = \lambda I$ bo'lsa, u holda $DX = \lambda X = XD$ har qanday X uchun.

15. $u = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$, $v = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$, $w = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$ **va** $L = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ **bo'lsin. Qirralarni qayta hisoblamasdan,** L **har bir qirraga qo'llanilgandan keyingi ishorali aralash ko'paytma va hajmni toping; hajm koeffitsienti nima uchun universal?**

$$\begin{bmatrix} 1 & 0 & 3 \\ 0 & 0 & -1 \end{bmatrix}$$

belgili aralash ko'paytma va hajmdan L **ni har bir qirraga qo'llaganidan keyin toping; nima uchun hajm koeffitsienti universal?**

[Manba: LA darsi, 3-hafta, masala 15]

Yechim. $U = \begin{bmatrix} u & v & w \end{bmatrix}$ deb olaylik; 12-masaladan, $\det U = 13$. O'zgargan qirralar LU ni hosil qiladi, shuning uchun $\det(LU) = \det L \det U = (-2)(13) = -26$; yangi hajm 26. $|\det L| = 2$ koeffitsienti U dan mustaqil, chunki $\det(LU) = \det L \det U$ har qanday U uchun — determinantlar ko'paytmasi qoidasi geometrik ma'noni anglatadi: $|\det L|$ — hajmni o'zgartirish koeffitsienti.

5 4-hafta — Matritsa rangi va teskari matritsa (Lineal algebra I)

Lineal algebra I kursining 4-haftasi matritsa rangi va Gauss-Jordan usuli bilan matritsani teskarisini olishni o'z ichiga oladi, 1-3 haftalar bo'yicha Test 1 bilan tugaydi. Repozitoriyda 4-hafta uchun alohida mashqlar varaqasi yo'q, shuning uchun bu qism (i)

Strangning berilgan materialidan to’plangan ikki mavzuga to’liq formulalar katalogi va (ii) TA kursi varaqalari uslubida gradatsiyalashgan mashq to’plamini to’liq yechimlari bilan beradi. Ular kursning boshqa joylarida bir xil matematik tuzilmalardan foydalanadi, shuning uchun bir vaqtda 1-testga matritsa-algebra qismlari bo’yicha tayyorgarlik xizmatini qiladi.

5.1 4-hafta formulalar katalogi

6.1.1 Rang va yo’q qilish

$$\operatorname{rank} A \le \min(m, n) \tag{106}$$

$$A \in \mathbb{R}^{n\times n}:\ \operatorname{rank} A = n \Leftrightarrow Ax = 0 \text{ faqat } x = 0 \text{ yechimga ega} \tag{107}$$

$$A \text{ teskarililigi} \Leftrightarrow \operatorname{rank} A = n \Leftrightarrow n \text{ ta pivot} \Leftrightarrow \det A \neq 0 \tag{108}$$

$$A \text{ teskarililigi} \Leftrightarrow Ax = b \text{ har bir } b \text{ uchun aynan bitta yechimga ega} \tag{109}$$

6.1.2 Teskari matritsa (Strang, Introduction to Linear Algebra 4e, §2.5)

$$AA^{-1} = I = A^{-1}A; \quad x = A^{-1}b \text{ bo‘lsa, } Ax = b \text{ yechim} \tag{110}$$

$$A^{-1} \text{ mavjud} \Leftrightarrow \text{yo‘q qilish } n \text{ ta pivot beradi (qator almashtirishlar mumkin)} \tag{111}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \quad (ad - bc \neq 0) \tag{112}$$

$$(AB)^{-1} = B^{-1}A^{-1}; \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \text{ (teskari tartib)} \tag{113}$$

$$\operatorname{diag}(d_1, \dots, d_n)^{-1} = \operatorname{diag}\left(\frac{1}{d_1}, \dots, \frac{1}{d_n}\right) \ (d_i \neq 0) \tag{114}$$

$$x \neq 0,\ Ax = 0 \Rightarrow A \text{ teskarili emas} \tag{115}$$

$$BA = I,\ AC = I \Rightarrow B = B(AC) = (BA)C = C \text{ (yagonalik)} \tag{116}$$

(115) ning izohi: hech bir matritsa 0 ni orqaga x ga qaytara olmaydi.

6.1.3 Gauss–Jordan yo’q qilish

$$[\,A \mid I\,] \xrightarrow{\text{boshlang‘ich qator amallari}} [\,I \mid A^{-1}\,] \tag{117}$$

Bir xil qator amallari A ni I ga va I ni A^{-1} ga aylantiradi. Qator amallari: ikkala qatorni almashtirish; qatorni noldan farqli konstantaga ko’paytirish; bitta qatorning ko’paytmasini boshqasiga qo’shish. Har biri teskari oson (amalni bekor qilish) bo’lgan boshlang’ich E matritsaga mos keladi, shuning uchun $(E_2E_1A)^{-1} = A^{-1}E_1^{-1}E_2^{-1}$ teskari tartibda chiqadi.

5.2 4-hafta mashqlari (rang, teskari, Gauss-Jordan) — savollar va to'liq yechimlar

1. $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 5 \\ 3 & 6 & 4 \end{bmatrix}$ **uchun chiqarib tashlash orqali** $\mathrm{rank}A$ **ni hisoblang.**

[Manba: LA 4-hafta amaliyoti, masala 1]

Yechim. Birinchi ustun elementi (1) ostida chiqaramiz: $R_2 - 2R_1 \to (0, 0, 3)$; $R_3 - 3R_1 \to (0, 0, 1)$. Ikkinchi ustun elementlari yo'qoldi, demak ikkinchi ustun elementi 3-ustundan keladi: ustun elementi $= 3$. Uning ostida chiqaramiz: $R_3 - \frac{1}{3}R_2 \to (0,\ 0,\ 0)$. Ikkita nol bo'lmagan ustun element: $\mathrm{rank}A = 2$. E'tibor bering $\mathrm{col}_2 = 2\mathrm{col}_1$, bu boshidan ko'rinib turardi — rang kamchiligi belgisi. 2 1

2. $B =$**, uchta usul bilan teskari ekanligini aniqlang: ustun elementlar, 4 2 determinant va nol bo'lmani nol vektor.**

[Manba: LAming 4-haftalik amaliyoti, 2-masala]

Yechim. (Asosiy)$R_2 - 2R_1 \to (0, 0)$: faqat bitta asosiy — noyob. (Determinant) ad − bc $= 2 \cdot 2 - 1 \cdot 4 = 0$ — (112) bo'yicha noyob. (Nol-vektor)$B(1,\ -2)^T = (2 - 2,\ 4 - 4)^T = 0$ — $Bx = 0$ tenglamaning nol bo'lmagan yechimi, demak (115) bo'yicha B teskarisiz. Uchala test ham mos keladi. 212×2 formulasi bo'yicha hisoblang va $AA^{-1} = I$ ekanligini tekshiring.

3. $A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$ **matritsasining teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 3-masala]

Yechim. $\det A = 6 - 5 = 1$, shuning uchun (112) bo'yicha $A - 1 = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix}$. Tekshirish: $A \cdot A - 1 = [2 \cdot 3 + 1 \cdot (-5),\ 2 \cdot (-1) + 1 \cdot 2;\ 5 \cdot 3 + 3 \cdot (-5),\ 5 \cdot (-1) + 3 \cdot 2] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$.

$$A^{-1} = (\frac{1}{7}) \cdot \begin{bmatrix} 2 & -3 \\ 1 & 5 \end{bmatrix}$$

$$A = [5 \cdot 3 + 3 \cdot (-5) 5 \cdot (-1) + 3 \cdot 2; 01] = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

4. $\begin{bmatrix} A & I \end{bmatrix}$ **matritsiga Gaus-Jordan usulini qo'llab** $A =$**. 3 7 matritsasini teskarisini toping.**

[Manba: LAming 4-haftalik amaliyoti, 4-masala]

Yechim.

$$\begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix} \to R_2 - 3R_1 \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \to R_1 - 2R_2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

$$A^{-1} = \begin{bmatrix} -7 & 2 \\ 3 & -1 \end{bmatrix}$$

Shunday qilib $A^{-1}=$; tekshirish: $AA^{-1}==$.

1

5. Gaus-Jordan usuli bilan teskarisini toping va tekshiring, 0 −1 2 asosiy ko‘paytmalari $\det K$ **ga teng ekanligini.**

[Manba: LAming 4-haftalik amaliyoti, 5-masala (Strangning sevimli misoli)]

Yechim. $[K|I]$ da ishlaymiz: birinchi opor 2 bilan chiqaramiz $(R2+(\frac{1}{2})R1)$, so’ngra ikkinchi opor $\frac{3}{2}(R3+(\frac{2}{3})R2)$, so’ngra uchinchi 4/3; yuqoriga davom etib, oporlar ustidini tozalaganda o’ngda $K-1$ hosil bo’ladi.

$$K^{-1}=(\frac{1}{4})\cdot\begin{bmatrix}3&2&1\\2&4&2\\1&2&3\end{bmatrix}$$

bo‘ladi. Asosiy ko‘paytmalar $2\cdot\frac{3}{2}\cdot 4_3=4=\det K$ ga teng, va haqiqatan ham K^{-1} butun sonli matritsani $\det K$ ga bo‘ladi — (112) dan keyin belgilangan namuna. (K simmetrik va uch diagonal, lekin K^{-1} zich.)

4-masaladagi A^{-1} yordamida.

6. $Ax=b$ **tenglamani** $A=, b=371$

uchun yeching.

[Manba: LAming 4-haftalik amaliyoti, 6-masala]

Yechim. $x=A{-}1\text{b}=\begin{bmatrix}7&-2\\-3&1\end{bmatrix}\cdot\begin{bmatrix}1\\1\end{bmatrix}=[7\cdot 1-2\cdot 1;\ -3\cdot 1+1\cdot 1]=\begin{bmatrix}5\\-2\end{bmatrix}.$

$$A^{-1}=(1)/(7)\begin{bmatrix}1&-2\\3&1\end{bmatrix}$$

; $(AB)^{-1}$ ni avval

7. $A=$ **va** $B=0211$ **matritsalar berilgan.** $(AB)^{-1}$ **ni avval**

[Manba: LAming 4-haftalik amaliyoti, 7-masala]

Yechim. (113) formulaga ko’ra, $(AB)^{-1}=B^{-1}A^{-1}$. Bu yerda $\det A=2$ va $\det B=2$, shuning uchun

$$det(AB)=4, teskarimatritsani(AB)^{-1}=(1)/(4)\begin{bmatrix}2&-3\\-2&4\end{bmatrix}$$

$$\begin{bmatrix}1&0\\0&1\end{bmatrix}[(-121)/2(21)/(-2);$$

$$(-314)/4(21)/(-2)]=[\frac{1}{2}\frac{1}{2};$$

$$\frac{3}{4}\frac{1}{4}]\text{—bir xil matritsa.}Va$$

$(AB) - 1(AB) = [\frac{1}{2}\frac{1}{2};\ \frac{3}{4}\frac{1}{4}]\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix} = [\frac{1}{2}\frac{1}{2};$ 3/4 1/4] * [(−1 2 1)/2 (2 1)/(−2); $(-314)/4(21)/(-2)] = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}$. Teskari tartib qonuni tasdiqlandi.

8. (Tushunchaviy, Test-1 uslubida.) To'g'ri yoki noto'g'ri, asoslari bilan: (a) **Agar** A **matritsa teskari bo'lsa,** A^T **ham teskari bo'ladi.** (b) **Agar** $Ax = \lambda xx \neq 0$ **va** $\lambda = 0$ **bo'lsa,** A **teskari emas.** (c) **Teskari** 3×3 **matritsalar ko'paytmasi noyob bo'lishi mumkin.** (d) **Har qanday** 2×3 **matritsa uchun** $\text{rank}(A) \leq 2$**.**

[Manba: LAming 4-haftalik amaliyoti, 8-masala]

Yechim. (a) To'g'ri: A da n ta asosli element bo'lsa $\Leftrightarrow$ A^T da n ta asosli element bo'ladi (qator rangi = ustun rangi); rasmiy ravishda $(A^T)^{-1} = (A^{-1})^T$. (b) To'g'ri: $Ax = 0 \cdot x = 0$ tenglamani qanoatlantiruvchi nol bo'lmagan $x(115)$ shartni buzadi. (c) Noto'g'ri: (113) formulaga ko'ra teskari matritsalar ko'paytmasi ham teskari bo'ladi. (d) To'g'ri: rang matritsa satrlari sonidan oshmaydi (har bir asosli element bitta satrni egallaydi), shuning uchun rank $\leq 2 < 3$.

9. (Amaliy, Test-1 uslubida.)c **ning qanday qiymat(lar)ida** $A = \begin{bmatrix}1 & c\\ 4 & 2\end{bmatrix}$ **matritsasi yagona (singulyar emas)? Qolgan qiymatlar uchun** $Ax = \begin{bmatrix}6\\ 4\end{bmatrix}$ **ni yeching.**

[Manba: LAming 4-haftalik amaliyoti, 9-masala]

Yechim. $\det A = c - 6$; noyob bo'lishi uchun aynan $c = 6$. Aks holda $A^{-1} =^1$

$$\det A = (c-6)/(4c-20)$$

vax $= A^{-1}b =^1$, masalan $c = 7$ da:$x = (8, -2)^T$.

$$x = A^{-1}b = (\text{begin}arrayc8 - 2endarray).$$

10. (Yuqori murakkablik.) Isbotlang, agar A **teskari bo'lsa,** u **holda** $\text{rank}(AB) = \text{rank}(B)$ **har qanday mos keladigan** B **uchun.**

[Manba: LAm 4-hafta amaliyoti, 10-masala]

Yechim. Teskari matritsaga ko'paytirish soni asosli vektorlarni o'zgartirmaydi: standavy fikrlash: $\text{rank}(AB) \leq \text{rank}B(AB$ qatorlari — B qatorlarining chiziqli kombinatsiyalari), va xuddi shuni $B = A^{-1}(AB)$ uchun qo'llaymiz, rank$B \leq \text{rank}(AB)$ hosil bo'ladi; ikkala tengsizlik tenglikni talab qiladi. Intuitsiya: teskari ko'paytuvchi faqat fazani cho'zadi/aylantiradi/siljitadi, lekin tekislamaydi, shuning uchun rang yo'qolishi yoki paydo bo'lishi mumkin emas.

6 Darsliklardan qo’shimchalar — to’liq yechimlari bilan majburiy o’qish uchun qo’shimcha mashqlar

6.1 Strang, Introduction to Linear Algebra 4e — tanlangan masalalar yechilgan

1. (§1.1) Barcha cu ni tavsiflang $u = (2, 1)$ uchun va barcha cu + dv kombinatsiyalarini $u = (2,\ 1),\ v = (1,\ 2)$ uchun; qachon cu + dv kombinatsiyalari tekislikni to’ldiradi?

[Manba: Strang Intro 4e, g’oyalar to’plami §1.1]

Yechim. cu (2, 1) orqali o’tgan to’g’ri chiziqni to’ldiradi: undagi har bir vektor — uning biror karrali. cu + dv koordinata boshidan o’tgan tekislikni to’ldiradi, agar $21u$ vav kollinear bo’lmasa; bu yerda det $= 3 \neq 0$, shuning uchun ikki yo’nalat 1 2 mustaqil va kombinatsiyalar butun R^2 ni to’ldiradi. Aynan shu mezon — nol bo’lmaydigan determinant — 4-haftada teskarilik sifatida qaytadi (108).

2. (§1.2) Skalyar ko’paytmalarni hisoblang:$u \cdot v$ uchunu $= (1,\ 2,\ 1), v = (3,\ 0,\ 2)$; ikkalasiga perpendikulyarw vektorni toping.

[Manba: Strang Intro 4e, masalar to’plami §1.2]

Yechim. $u \cdot v = 3 + 0 + 2 = 5$. Perpendikulyar vektor — vektor ko’paytmasi:$w = u \times$v =

$$\begin{bmatrix}1\\2\\1\end{bmatrix} \times \begin{bmatrix}3\\0\\2\end{bmatrix} = [(2 \cdot 2 - 1 \cdot 0);$$

$$(1 \cdot 3 - 1 \cdot 2); (1 \cdot 0 - 2 \cdot 3)] = \begin{bmatrix}4\\1\\-6\end{bmatrix}$$

; tekshirish: $w \cdot u = 4 + 2 - 6 = 0$ va $w \cdot v = 12 + 0 - 12 = 0$.

3. (§1.2) Qachonu $\cdot v = v \cdot u$ manfiy va bu burchak haqida nima deydi? Nolga teng qachon?

[Manba: Strang Intro 4e, tushuncha §1.2]

Yechim. Manfiy aynan $\cos\theta < 0$ bo’lganda, ya’ni burchak 90° dan katta (o’tkir); nol $\theta = 90^\circ$ da — perpendikulyarlik. Bu tasnif, 2-haftadagi TA masalarida qo’llanilgan. $21, b = (3, 4)^T$ orqali A^{-1}.

4. $(§2.5) Ax = b$ **ni yeching** $A = 12$

$$[2 - 12; 3 - 42; \frac{2}{3}]$$

[Manba: Strang Intro 4e, masalar to’plami §2.5]

Yechim. $\det A = 3$: $A^{-1} =^1$;$x = A^{-1}b =^1 =^1 =$.

$$\det A = 3; A = [1 - 12; 3 - 38; 35\frac{5}{3}]; x = A^{-1}b = [\frac{1}{4}; 4;$$

$$2] = [\frac{14}{4}; 4; 2] = [\frac{7}{2}; 4; 2](16)$$

Tekshirish: $2\cdot(\frac{2}{3})+5\cdot(\frac{1}{3})=3$, $2\cdot(\frac{1}{3})+2\cdot(\frac{5}{3})=4$.

5. (§2.5) Nol bo'lmagan $Ax=0$ yechimi teskarini istisno qiladi? Ikkita jumladan iborat fikrni keltiring, konsept «hech qanday matritsa0 nix ga qaytara olmaydi» deb nomlanadi.

[Manba: Strang Intro 4e, §2.5ga 4-e'lon]

Yechim. Agar $Ax=0x\neq 0$ va A da teskari bo'lsa, u holdax $=A^{-1}Ax=A^{-1}0=0$ — ziddiyat. Nol vektorning «teskari tasviri» faqat aynilikda mavjud; nol-vektor buni emasligini isbotlaydi.

6. (§2.5) Agar A va C teskari 3×3 matritsalar bo'lsa, $A+C$ teskari bo'ladimi? AC? $A^{-1}C$?

[Manba: Strang Intro 4e, masalar to'plami §2.5]

Yechim. $A+C$: albatta emas (masalan, $C=-A0$ beradi); teskari matritsalarning yig'indisi aynan bo'lishi mumkin. AC: ha, $(AC)^{-1}=C^{-1}A^{-1}(113)$. $A^{-1}C$: ha, ikkita teskari matritsaning ko'paytmasi.

6.2 Strang, Linear Algebra and its Applications 4e, §1.6 (Teskari va transponirlangan) — asosiy natijalar va tahlil qilingan misol

Belgilangan o'qish §1.6 katalogga uchta qo'shimcha element qo'shadi: Gauss– 2 1 1

$$\det A^{-1}=(1)/(\det A)$$

Jordani bilan $A=4-60_{detA}$ va transponirlanish -2 7 2 identifikatorlari. 2 1 1

1. $A=\begin{bmatrix}2&1&1\\4&-6&0\\-2&7&2\end{bmatrix}$ **uchun Gauss-Jordan usulini bajaring va $A-1$ ni ko'rsating.**

[Manba: Strang LAiA 4e, §1.6, misol 1]

Yechim. I ni to'ldirib, chiqarib tashlaymiz: asosiy $2,\,-8,\,1(R_2-2R_1,\,R_3+R_1$, keyin $R_3+\frac{3}{4}R_2$ amallaridan so'ng). Yuqoriga davom etib, asosiy elementlar ustini tozalab, satrlarni 2, -8, 1 ga bo'lib,

$$A^{-1}=(\frac{1}{8})\begin{bmatrix}12&-5&-6\\8&-3&-2\\-1&1&1\end{bmatrix}$$

hosil qilamiz. Determinant asosiy elementlar ko'paytmasiga teng $2(-8)(1)=-16$, va haqiqatan ham $A^{-1}\det A$ ga bo'lishni ko'rsatadi (kofaktorlarning belgilari hisobga olingan — $A^{-1}=\frac{adjA}{detA}$ ga ekvivalent).

2. §1.6 da qo'llanilgan determinant-transponirliqlik identifikatsiyalarini ko'rsating: $\det A^T$, $\det(AB)$, $\det A^{-1}$**.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $\det A^T=\det A$; $\det(AB)=\det A\det B(1-$jadval); va $B=A^{-1}$ deb olib, $\det(A)\det(A^{-1})=\det I=1$, shuning uchun $1\det A^{-1}=\det A$, bu esa "teskari mavjud $\Leftrightarrow\Rightarrow$ $\det A\neq 0$" degani aniq, determinantlar katalogini (3-hafta) teskari matritsalar katalogi (4-hafta) bilan bog'laydi.

3. Isbotlang: $(A^{-1})^T = (A^T)^{-1}$ **va tushuntiring, nima uchun** A^TA **simmetrik va manfiy bo'lmagan diagonalga ega.**

[Manba: Strang LAiA 4e, §1.6]

Yechim. $AA^{-1} = I$ ni transponirib, $(A^{-1})^T A^T = I$ hosil bo'ladi, bu esa

$$(A^{-1})^T = (A^T)^{-1} \quad (16)$$

degani aniq: $(A^{-1})^T$ — A^T ning teskarisi.

$$(A^T)^{-1} = (A^{-1})^T; \text{matritsa} A^T A \text{simmetrik}:$$
$$(A^TA)^T = A^TA$$

uning diagonal elementlari manfiy emas :

$$a_{ii} = \Sigma_{k=1}^{n} a_{ki}^2 \geq 0 \text{—kvadratlar yig}'\text{indisi — fakt.} \quad (17)$$